MILITARY SPECIFICATION

CONTROL, CENTRAL C-3076 (*)/ASQ-19

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide control of the UHF Communication, UHF Navigation, UHF Aux Receiver, and UHF ADF equipment portions of the Integrated Electronic Central AN/ASQ-19. It shall also provide control information for two Indicators Frequency Channel ID-808/ASQ.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-18309	Integrated Electronic Central AN/ASQ-19(*)
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment
MIL-C-6781	Control Panel, Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Lighting

STANDARDS

Federal

FED-STD-3	Colors, Aeronautical Lighting

2.2 Availability of Documents - When requesting specifications, publications, and drawings refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions, may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

FSC 5841

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3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, parts and material requirements shall be in accordance with Specification MIL-I-18309.

3.2 Design and Construction

3.2.1 General - This equipment shall conform with all the applicable requirements of Specification MIL-I-18309 for design, construction and workmanship except in the paragraphs as modified below:

3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 7-1/8 pounds.

3.2.4.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.5.1 Cables and Connectors - The external connectors shall be one each Bendix Type PT01E-22-55P, PT01E-55PW, and PT01E-18-32P or approved equivalents mounted on pendant cables at the rear of the equipment.

3.2.11 Control Panels - All rack or console mounted panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to the preproduction testing.

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 UHF Communication Portion of C-3076/ASQ-19

3.3.2.1 Frequency and Channel Selection - Manual and preset frequency selection is provided for in the range 225.0 to 399.9 mc. Provision is made for dual remote channel indicators to be used in conjunction with this equipment.

3.3.2.2 Volume Control - Operation of the volume control shall cause a minimum of change in the impedance which the radio set presents to the external equipment attached to the audio output terminals, and such impedance shall not be less than 50 ohms at any setting of the volume control.

3.3.2.3 Function Control - Operation of the function selector control shall permit selection of standby operation, selection of transmission and reception on the main frequency, selection of transmission and reception on the main frequency and reception on the guard frequency, selection of automatic direction finding on the main frequency and reception on the guard frequency, and selection of automatic direction finding on the main frequency. This control shall operate in conjunction with the UHF Auxiliary Receiver, and UHF ADF selector controls (see 3.4.1.3.1).

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3.3.3 UHF Auxiliary Receiver Portion of C-3076/ASQ-19

3.3.3.1 Channel Selection - Channel selection of any one of the twenty (20) available channels.

3.3.3.2 Volume Control - Operation of the volume control shall cause a minimum of change in the impedance which the receiver presents to the external equipment attached to the audio output terminals, and such impedance shall not be less than 50 ohms at any setting of the volume control.

3.3.3.3 Function Control - Operation of the function selector control shall permit selection of standby operation, selection of automatic direction finding operation, selection of reception on one of the twenty preset or command channels, and selection of reception on the guard frequency. This control shall operate in conjunction with the UHF Communication and UHF ADF selector controls (see 3.4.1.3.1).

3.3.4 UHF Navigation Portion of the C-3076/ASQ-19

3.3.4.1 Channel Selection - Channel selection on any one of the 126 available channels.

3.3.4.2 Volume Control - Operation of the volume control shall cause a minimum of change in the impedance which the radio set presents to the external equipment attached to the audio output terminals, and such impedance shall not be less than 50 ohms at any setting of the volume control.

3.3.4.3 Function Control - Operation of the function control shall permit selection of standby operation, selection of reception on the navigation channel, and selection of transmission and reception on the navigation channel.

3.3.5 UHF ADF Portion of the C-3076/ASQ-19

3.3.5.1 Function Control - Operation of the function selector control shall permit selection of standby operation and selection of automatic direction finding operation. This control shall operate in conjunction with the UHF Communication and UHF Auxiliary Receiver selector controls (see 3.4.1.3.1).

3.3.6 Dual Operation - Operation of the COMM CMD and NAV CMD controls shall permit either operator to "take control" separately over the navigation and communication functions.

3.3.7 Audio Circuits - The volume controls used in the equipment shall be attenuators, which for proper operation, require that the output terminal of the attenuator be connected to suitable loads. Loads suitable for this purpose may be contained within the equipment, but their connection to the audio circuits shall be at the discretion of the user and shall be dependent upon the over-all audio load.

3.4 Detail Requirements

3.4.1 The equipment shall be of the console type in accordance with Specification MIL-C-6781, unless otherwise specified herein, and shall fulfill the following requirements.

3.4.1.1 Function - The equipment shall provide control of the UHF Communication, UHF Auxiliary Receiver, UHF Navigation, and UHF ADF equipment portions of the Integrated Electronic Central AN/ASQ-19. It shall be possible to select manually any one of the 1750 frequencies available on the UHF Communication portion of the AN/ASQ-19 and to preset up to 18 channels on any one of these available frequencies. Additionally, it shall be possible to select a fixed tuned guard channel. Unless otherwise specified, the equipment shall be supplied with the guard frequency set for 243.0 mc. The equipment shall also provide for dual remote indication of the UHF Communications preset channels (1 through 18), guard channel and manual operation. Provisions shall be available for dual

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operation of the UHF Communication, UHF Auxiliary Receiver, and UHF Navigation portions of the AN/ASQ-19. It shall be possible also to select any one of the twenty (20) UHF Auxiliary Receiver channels and to select operation of the UHF ADF equipment when desired. Complete function and channel selection of the UHF Navigation shall be provided also.

3.4.1.2 Form Factor - The dimensions of the Central Control C-3076/ASQ-19 shall conform to the applicable requirements of Specification MIL-C-6781 for a Type I control panel and shall not exceed 8-5/8 inches in length and 5-5/8 inches in depth. Projecting knobs shall not exceed 1-1/4 inches from the mounting surface.

3.4.1.3 Controls, Panel Mounted - The following controls shall be located and suitably marked on the front panel.

3.4.1.3.1 UHF COMM, UHF AUX REC

(1) Three (3) controls (COMM FREQ MC) for manually selecting any one of the 1750 available frequencies on the UHF COMM.

(2) A twenty (20) position rotary switch-type control (COMM CHAN) for selecting any one of 18 predetermined frequencies on the UHF COMM, or the guard frequency, or a manual channel on which any of the 1750 available frequencies can be selected. Also an indication of the UHF Communication preset channels (1 through 18), manual or guard-frequency display for two remote indicators.

(3) A rotary volume control (COMM VOL) for adjusting the amplitude of the audio signals delivered to the output circuit by the UHF COMM.

(4) A five (5) position rotary-type switch for selecting the mode of operation as follows:

POSITION	UHF COMM	AUXILIARY RECEIVER
1	STBY	STBY
2	T/R	ADF
3	T/R+G	ADF
4	ADF + G	CMD
5	ADF	GRD

(5) A twenty (20) position rotary switch-type control (AUX CHAN) for selecting any one of the 20 UHF AUXILIARY RECEIVER channels.

(6) A rotary volume control (AUX VOL) for adjusting the amplitude of the audio signals delivered to the output circuit by the UHF AUXILIARY RECEIVER.

3.4.1.3.2 UHF NAV

(1) Two (2) controls (NAV CHAN) for manually selecting any one of the 126 NAV channels.

(2) A rotary volume control (NAV VOL) for adjusting the amplitude of the audio signals delivered to the output circuit by the NAV.

(3) A three (3) position rotary switch for selecting the mode of operation as follows: Position one-STBY; position two - REC operation on channel selected by (1) above; position three - T/R operation on channel selected by (1) above.

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3.4.1.3.3 SPECIAL

(1) A push-type "Take Control" switch for COMM CMD operation when dual controls are installed. This switch controls an external relay (or relays as necessary) to switch control functions as desired. An integral light in the push button is controlled by the external relay (or relays) and will be illuminated on the operator's panel that has control. This control also functions as a "Give" control if desired. Color of push button shall be identification green conforming to FED-STD-3.

(2) A push-type "Take Control" switch for NAV CMD operation identical in operational characteristics to (1) above.

(3) Emergency power (EMER POWER) indicator light to give warning when main power failure has occurred and equipment is operating on emergency power. Color of indicator light shall be identification yellow conforming to FED-STD-3.

3.4.1.4 Controls, Internal - Provisions shall be made for changing UHF COMM preset frequencies on the memory drum by removing unit from normal installation. An access cover shall be provided at the rear of the unit which will enable the operator to readily change preset frequencies.

3.4.1.5 Audio Loads - Audio load resistors having the characteristics listed below shall be connected from each of the audio output terminals of the equipment to a separate pin on the connector, where they may be individually grounded at the option of the user.

COMM VOL	620 ohms 1 watt
AUX VOL	620 ohms 1 watt
NAV VOL	1200 ohms 1 watt

3.4.1.6 Power Requirements - The equipment shall give specified performance with the 28-volt d-c power source listed in Specification MIL-I-18309. The d-c power required shall not exceed 10 watts.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the test requirements of Specification MIL-I-18309 to determine compliance with all applicable requirements except in the paragraphs as modified below:

4.4.1.3 Manufacturing Reliability Test - At the completion of the specified vibration period, each control of the equipment shall be operated for ten complete operations without failure.

4.4.2.1(3) Manufacturing reliability test specified in 4.4.1.3, except that the test shall consist of 200 operations of each control with no restrictions on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.5 Life Test - Life tests of the equipment shall consist of 10,000 operations of each of the COMM manual tuning controls and 25,000 operations of the COMM preset channel control and the AUX Receiver, NAV Coarse and NAV Fine channel selectors.

5. PREPARATION FOR DELIVERY

5.1 General - Preparation for delivery shall be in accordance with Specification MIL-I-18309.

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6. NOTES

6.1 Use - This equipment is mainly intended to be used to provide control of the communication and navigation functions of the Integrated Electronic Central AN/ASQ-19.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 volt d-c. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply.

(1) Contract - The contract shall have precedence over any specification.

(2) Central Specification - Integrated Electronic Central AN/ASQ-19 Specification MIL-I-18309 shall have precedence over this specification.

(3) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the procuring activity.

(4) Referenced Specifications - Any referenced specification shall have precedence over all applicable specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.10 of Specification MIL-I-18309. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

PATENT NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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